

```
import processing.serial.*;

Serial myPort;

float bpm = 0;

float[] graph = new float[300];

void setup() {

    size(1000, 800);

    println(Serial.list());

    myPort = new Serial(this, Serial.list()[1], 115200);
    myPort.bufferUntil('\n');

    for (int i = 0; i < graph.length; i++) {
        graph[i] = 70;
    }
}

void draw() {

    background(0);

    // small BPM text
    fill(255);
    textSize(20);
    textAlign(LEFT);

    if (bpm > 0) {
        text(int(bpm) + " BPM", 20, 30);
    } else {
        text("PLACE FINGER", 20, 30);
    }

    // graph
    stroke(255);
    strokeWeight(4);
    noFill();

    beginShape();

    for (int i = 0; i < graph.length; i++) {

        float x = map(i, 0, graph.length-1, 50, width-50);

        float y = map(graph[i], 40, 180,
            height-80,
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        180);

    vertex(x, y);
}

endShape();

// moving dot
float lastY = map(graph[graph.length-1],
    40, 180,
    height-80,
    180);

noStroke();
fill(255);
ellipse(width-50, lastY, 12, 12);
}

void serialEvent(Serial myPort) {

    String data = myPort.readStringUntil('\n');

    if (data != null) {

        data = trim(data);

        try {

            bpm = float(data);

            for (int i = 0; i < graph.length-1; i++) {
                graph[i] = graph[i+1];
            }

            graph[graph.length-1] = bpm;

        } catch(Exception e) {
            println("Bad Data");
        }
    }
}

```